Kasi Yedumati · Full Stack Training

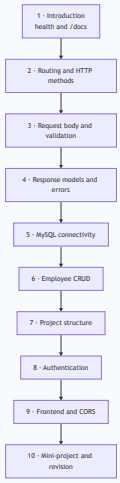
FastAPI for Full Stack — Learning Path

10 sessions × 90 minutes · Project: **AcmeHR** · Comes **after** HTML and MySQL, **before** Django

This module teaches FastAPI by building one API: the AcmeHR employee service. Every session adds to the same repository, `acmehr-api`. Django, taught next, is the full website. FastAPI is how that site's pages will ask for data.

Open a lesson here if you can write a small Python function, use VS Code, and run a command in the terminal. HTTP and JSON are explained in the lessons.

The path



#	Lesson	AcmeHR piece you build	Lab
1	Introduction to FastAPI	`GET /health` and `/docs`	L01
2	Routing and HTTP methods	Retrieve an employee by id and by department	L02
3	Request body and validation	Employee registration with checks	L03
4	Response models and error handling	A clean employee response, and a 404	L04
5	Database connectivity	Employee rows in MySQL, read through a route, a service, and a repository	L05
6	CRUD	Create, read, update, and delete employees	L06
7	Project structure	`routers/`, `services/`, `repositories/`, `Depends`, `.env`, one test	L07
8	Authentication	A token on a protected employee route	L08

#	Lesson	AcmeHR piece you build	Lab
9	Frontend and CORS	The registration page calls the API	L09
10	Mini-project and revision	Demo the Employee Management API	L10

Lessons 1–10 are ready to teach.

One repository

Keep one folder, `acmehr-api`, from the first health check through the demo. Homework finishes the current slice, then one 15-minute transfer drill: the same endpoint with different nouns (a book, a patient).

The people in the examples are the AcmeHR sample staff from the HTML portal:

Employee ID	Name	Designation	Department
E-1042	Ananya Rao	Senior Software Engineer	Engineering
E-1043	Karthik Menon	Engineering Manager	Engineering
E-1044	Meera Iyer	Accountant	Finance
E-1045	Rahul Sharma	HR Executive	Human Resources
E-1046	Fatima Khan	Product Designer	Product
E-1047	Arjun Reddy	QA Engineer	Engineering

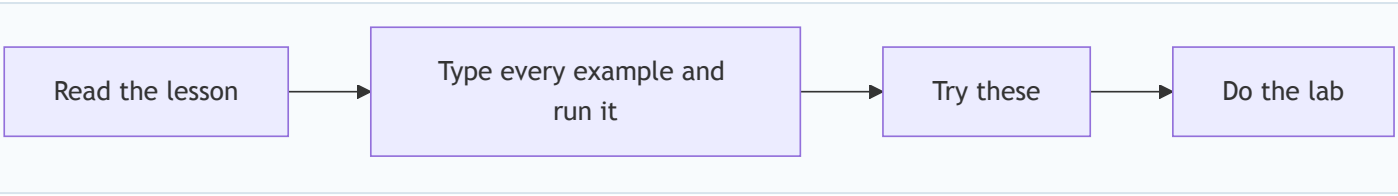
Ananya's manager is Karthik. The office in the samples is Hyderabad.

How a 90-minute session runs

Minutes	Block
15	Look at the picture for this lesson, and the few sentences under it
25	Trainer types the example. Learners type it too
30	Lab
10	Fix what broke
10	Homework: this slice, plus the transfer drill

How to study a lesson on your own

About 90 minutes.



- Type the code. Running a pasted file hides the mistakes you need to see.
- Open `/docs` after every change and try the endpoint there.
- Do the lab at the end of the lesson. **Lab** and **Homework** are folded. **Expand** opens that section. **Collapse** closes it. The PDF prints both in full.
- The code for that lab is the matching folder under `labs/`. The lesson prints the path. Open `solution/` only after the checklist passes.

Part	What it gives you
You will build	The AcmeHR slice for this session
New words	The words this lesson adds
Picture	The request you are about to make real
Code, then the screen	The program, then the terminal line or the JSON
Common error	The message, the cause, and the fix
Lab	A ticket from the Acme HR team

Setup (once)

- **Python:** 3.11 or 3.12
- **Editor:** VS Code
- **Browser:** Google Chrome
- **First API package:** `fastapi[standard]` — the `fastapi dev` command, which starts Uvicorn

```
mkdir acmehr-api
cd acmehr-api
python3 -m venv venv
source venv/bin/activate          # Windows: venv\Scripts\activate
pip install "fastapi[standard]"
```

From the database connection onward, the database name is `training_db`, the same one used in the MySQL module.